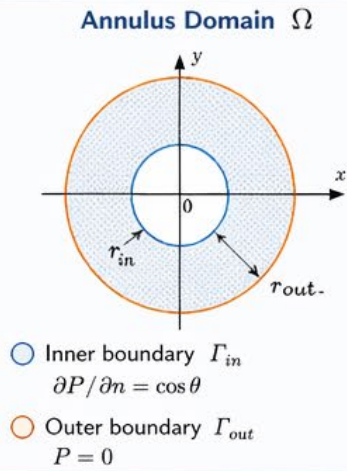


# 1 Physics & Geometry



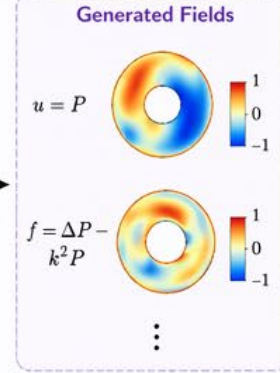
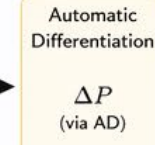
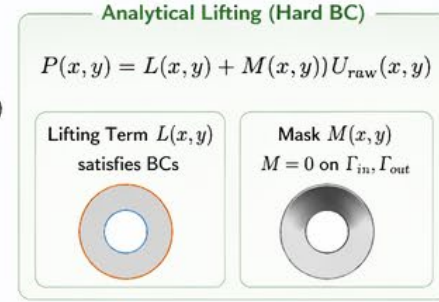
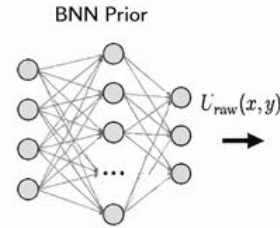
**PDE**

$$\Delta P - k^2 P = f \text{ in } \Omega$$

**Parameter**

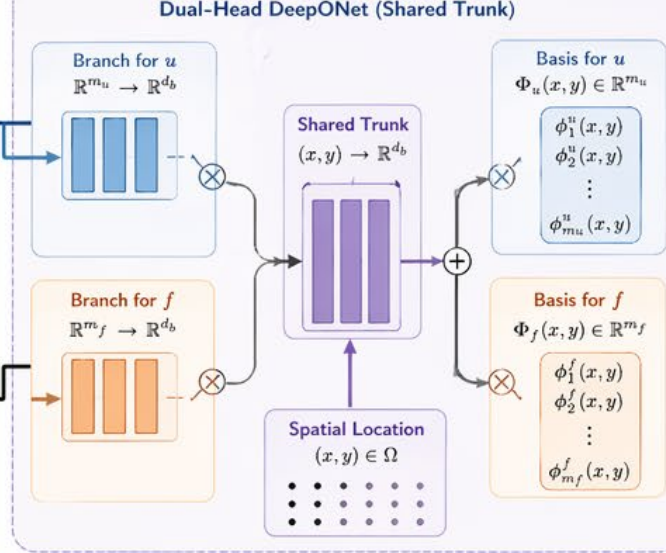
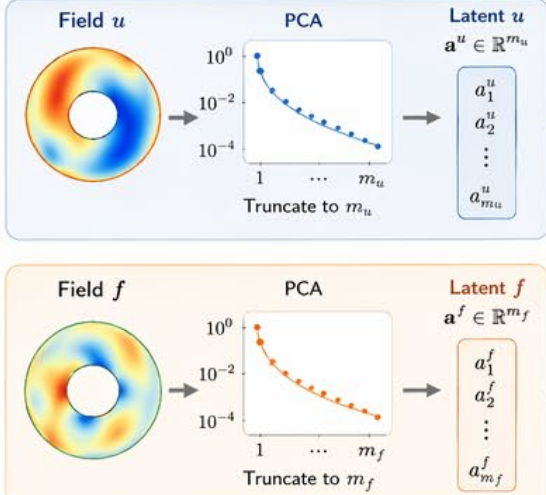
$$k \in \mathcal{D}_k$$

# 2 PI-Sampler (Physics-Informed Sampler)

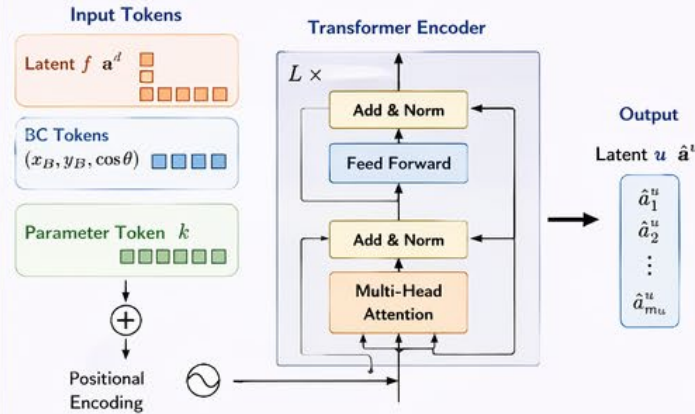


Sample many  $(u, f)$  pairs over  $\Omega$  with varying  $k$

# 3 Function Encoder (Representation Learning)



# 4 Operator Learning (Transformer)



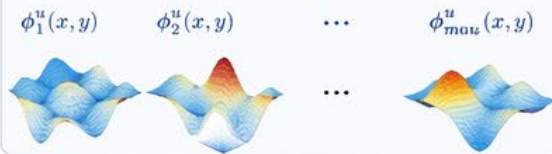
# 5 Reconstruction

Predicted Latent  $\hat{\mathbf{a}}^u$

$\hat{a}_1^u, \hat{a}_2^u, \dots, \hat{a}_{m_u}^u$

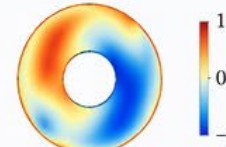
$\times$

**Learned Basis  $\Phi_u(x, y)$**



$=$

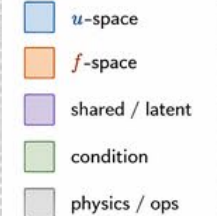
**Reconstructed Field  $\hat{u}(x, y)$**



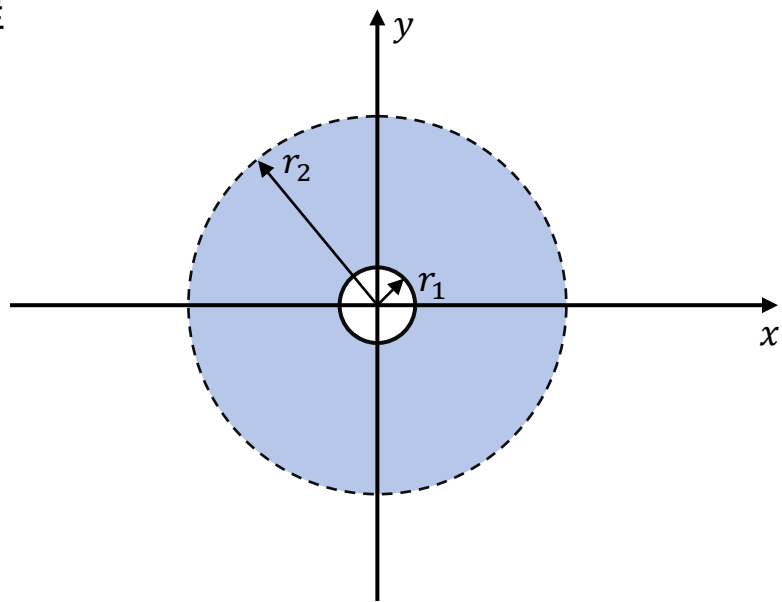
**Training Objective (Self-Supervised)**

Minimize reconstruction error of  $u$  at probe points  $\mathcal{X}_{probe}$

$$\mathcal{L} = \|u_{true} - \hat{u}\|_{\mathcal{X}_{probe}}^2$$



## 方程



$$\begin{cases} \nabla^2 P - k^2 P = f \\ \frac{\partial P}{\partial r} = -\cos(\theta), \quad r = 1 \\ P = 0, \quad r = 5 \end{cases}$$

硬编码边界条件:  $P_{BNN} = -(r - 5)\cos(\theta) + (r - 5)(r - 1)^2 U_{BNN}$

## config

```
# Geometry
r_inner: float = 1.0
r_outer: float = 5.0

# PDE parameter range
k_min: float = 0.05
k_max: float = 1.5

# PI-sampler prior
dim: int = 2
hidden_bnn: int = 256
sigma_list: tuple[float, ...] = (3.0, 5.0, 7.0)
sample_size: int = 256
```

```
# Sampling and discretization
theta_size: int = 128
radial_size: int = 32
random_probe_points: int = 1024
pod_snapshots: int = 200
n_basis: int = 512
```

```
# Function encoder
branch_width: int = 256
branch_depth: int = 4
trunk_width: int = 256
trunk_depth: int = 4
```

```
# Transformer
transformer_dim: int = 256
transformer_heads: int = 8
transformer_layers: int = 4
transformer_mlp_dim: int = 512
seq_chunks: int = 32
cond_chunks: int = 32
```

```
# Training
fe_b1 = 0.5
fe_b2 = 0.9
fe_steps: int = 100_000
ol_steps: int = 150_000
fe_lr: float = 1e-3
ol_lr: float = 1e-3
weight_decay: float = 1e-6
seed: int = 0
```

```
# Sampling and discretization
cfg.n_basis = 2048
cfg.theta_size = 128
cfg.radial_size = 32
cfg.pod_snapshots = 200

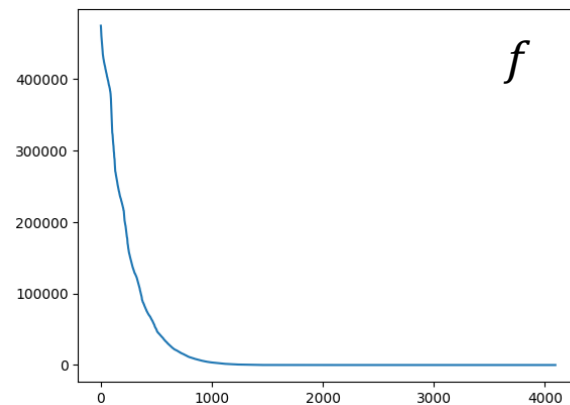
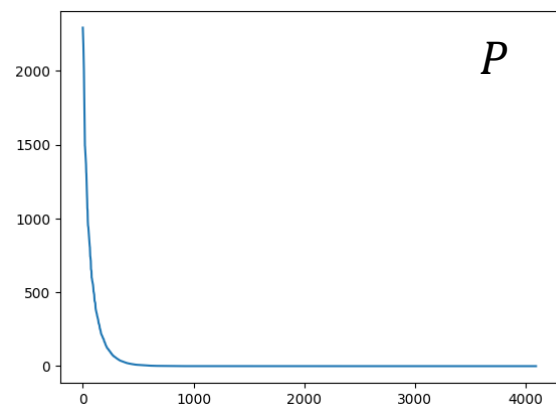
# PDE parameter range
cfg.k_min = 0.02
cfg.k_max = 0.02

# PI-sampler prior
cfg.sigma_list = (3.0, 3.0, 3.0)

# Function encoder
cfg.branch_depth = 6
cfg.trunk_depth = 6
```

问题1: PCA  $P$  和  $f$  所需要的基函数不同

目前我是分别对  $P$  和  $f$  做 PCA 得到各自的系数, 方差衰减如图。



方差衰减不一致, 需要用不同维度的PCA系数分别表示。

目前我设置在 Branch net 里设置两个 head, 分别接收  $P$  和  $f$ , 再融入同一个 Branch body 得到POD系数。

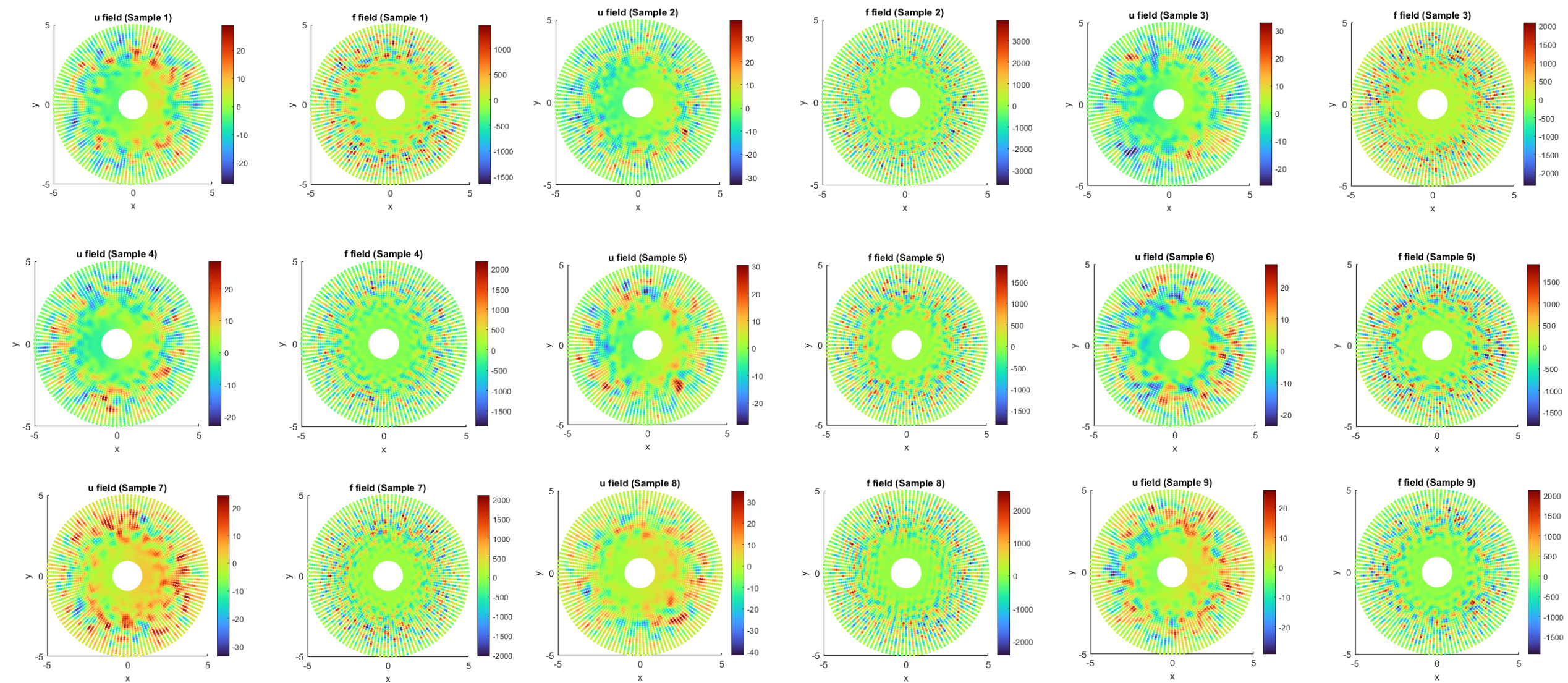
需不需要换成卷积? 需要融入一个Branch body吗? 还是设置两个 Branch net 使其共用一个 Trunk net?

问题2: 生成的  $P$  和  $f$  有比较显著的差异

导出查看用BNN构造的  $P$  和  $f$  的分布, 发现两者在频域和值域上都有比较显著的差异, 是否还能用同一组基函数表示?

选取10组分布图, 下页展示。





可以看出  $P$ (图中的  $u$ ) 和  $f$  的差异很大,  $f$  偏向高频波动且波动范围剧烈。尝试过通过将  $P$  的值域进行放缩, 在训练时有一定的效果, 但是效果不明显。